

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Tech. (CE/IT/EC) Examination

April 2013

ME102 Fundamentals of Mechanical Engineering

Date: 02.05.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q – 1 (a) What is calorific value? Explain types of calorific value. [04]
- (b) Find out the Work done, Heat transfer and Change in Internal energy for Constant Pressure Process. [03]
- Q - 2 (a) Determine enthalpy and internal energy of 1 kg of steam at a pressure of 12 bar when (i) the dryness fraction of steam is 0.8 (ii) steam is dry and saturated (iii) steam is superheated to 280°C. Take $C_{ps} = 201 \text{ kJ/kg K}$. Use steam table. [07]
- (b) Explain Cochran Boiler with its Construction & working & all notations with neat sketch. [07]

OR

- Q – 2 (a) Explain Construction and working of Combined separating and throttling calorimeter with neat sketch and also calculate the equation of dryness fraction. [07]
- (b) The following observation were made in boiler trial of 5 hours duration. [07]
Steam pressure: 11 bar, Dryness fraction of steam: 0.90, Mass of steam generated: 35,000 kg, Feed water temperature: 30°C, Coal supplied to boiler: 3600 kg, Calorific value of coal: 32,800 kJ/kg. Assume $C_{pw} = 4.187 \text{ kJ/kg K}$. By using steam table calculate:
- a. Evaporation rate
 - b. Generation factor (f_{ge})
 - c. Heat rate of boiler
 - d. Equivalent evaporation
 - e. Boiler efficiency
 - f. Boiler power

- Q - 3 (a) Classify Pumps. Explain Centrifugal pump with neat sketch and mention the function of all its parts. [07]
- (b) Explain working processes of Vapour compression refrigeration cycle with neat sketch. [07]

OR

- Q - 3 (a) Write down the comparison between belt drive, chain drive & gear drive. [07]
- (b) Explain working processes of Air refrigerator. [03]
- (c) Differentiate between brake & Clutch [04]

SECTION – II

- Q - 4 (a) What is a thermodynamic system? Explain Zeroth law of thermodynamics. [03]
(b) State second law of thermodynamic and give the difference between Extensive and Intensive properties. [04]
- Q - 5 (a) Explain Spark Ignition Four stroke engine system with p-V diagram and neat sketch. [07]
(b) Draw P-V diagrams for constant volume heat addition and constant pressure heat addition processes with all notations. [04]
(c) Explain relation between C_p and C_v . [03]

OR

- Q - 5 (a) One kg of air at a pressure of 1 bar temperature of 27°C is compressed to a pressure of 10 bar isothermally and adiabatically. Calculate work done, heat transfer and change in internal energy in both cases. Take $C_p = 1.005 \text{ kJ/kg K}$, $C_v = 0.718 \text{ kJ/kg K}$ [07]
(b) Explain work required for compression without clearance for single stage compressor. [07]
- Q - 6 (a) Explain construction and working of reciprocating compressor with labeled diagram. [07]
(b) What is an isentropic process and prove that $PV^\gamma = C$. [04]
(c) Define following terms: [03]
1. Prime mover
2. Isobaric process
3. Enthalpy

OR

- Q - 6 (a) Draw the figure of I.C. Engine & give its classification. [07]
(b) The following readings were taken during the test of single cylinder 4-stroke engine. [07]
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|---------------------------------|----------------|
| Cylinder diameter | = 270 mm |
| Stroke length | = 380 mm |
| Mean effective pressure | = 6 bar |
| Engine speed | = 250 rpm |
| Neat load on the break | = 1000 N |
| Effective diameter of the break | = 1.5 m |
| Fuel used per hour | = 10 kg/hr |
| Calorific value of fuel | = 44,400 kJ/kg |
- Calculation:
- Indicated power
 - Break power
 - Mechanical efficiency
 - Indicated thermal efficiency
 - Brake thermal efficiency
 - Brake specific fuel consumption
